1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20

Customer:	SOUND ENERGY
Contractor:	I.M.S.P. MOROCCO
Engineering:	ITALFLUID
Technology vendor:	TECHNICAL AMERICA INC

Title: **LNG TENDRARA PLANT**

objet: **P-PN-OS412.1**

HOT OIL CIRCULATION PUMP SKID

Customer code: -	Sheets : 16
By: -	S/N. : IE25072_SE04
File : -	

esaelettromeccanica.com

INGEGNERIA ELETTRICA - QUADRI DI DISTRIBUZIONE E AUTOMAZIONE - CABINE DI TRASFORMAZIONE MT/BT – PLC – HMI – SCADA - ENERGIE RINNOVABILI

CONTRADA TESINO, 47 - 63065 RIPATRANSONE (AP) – ITALY - TEL. +39 0735 90523

WWW.ESAELETTROMECCANICA.COM - INFO@ESAELETTROMECCANICA.COM - ESA@PEC.IT - P.IVA e C.F. 00768110447

General Contractor :	Manufacture : esa	Project : P–PN–OS412.1 HOT OIL CIRCULATION PUMP SKID	Customer : SOUND ENERGY	Object :	By :
			Contract : IE25072	Rev. 3 Data : 07/07/2025	Date : 17/02/2025
			S/N : IE25072_SE04	Rev. 3 Data : 17/04/2025	Sheet : 1 of : 16

CIRCUIT SYMBOLS ACCORDING TO CEI STANDARD

	ISOLATOR
	FUSED ISOLATOR
	CIRCUIT BREAKER WITH THERMAL AED MAGNETIC OVERCURRENT RELEASES
	CIRCUIT BREAKER WITH MAGNETIC RELEASE
Id	CIRCUIT BREAKER WITH THERMAL, MAGNETIC AND DIFFERENTIAL RELEASES
Id	RESIDUAL CURRENT CIRCUIT BREAKER "AC" TYPE FOR ALTERNATING CURRENT "A" TYPE FOR ALTERNATING OR PULSATING CURRENTS
	THERMAL RELAY (OVERLOAD RELAY)
	COIL OF CONTACTOR RELAY
BA	COIL OF SHUT-OFF ALARM MAKE CONTACT ON ENERGIZATION
U<	COIL OF SHUT-OFF ALARM DENERGIZES IN CASE OF UNDERVOLTAGE
	COIL OF OPERATING DEVICE WITH DROP-OUT DELAY
	COIL OF OPERATING DEVICE WITH PICK-UP DELAY
	COIL OF OPERATING DEVICE WITH DROP-OUT AND PICK-UP DELAYS
	COIL OF LATCHING RELAY (STEP-BY-STEP FUNCTION)
	ELECTRONIC OR ELECTRO-MECHANICAL TIME SWITCH

	ELECTRONIC TIMER WITH PICK-UP DELAY
	ELECTRIC PULSE METER
	FUSE CARRIER
	MAKE CONTACT OF CONTACTOR
	MAKE CONTACT OF CONTACTOR
	MAKE AND BREAK CONTACT
	MAKE AND BREAK CONTACT WITH DROP-OUT DELAY
	MAKE AND BREAK CONTACT WITH PICK-UP DELAY
	MAKE AND BREAK CONTACT WITH DROP-OUT AND PICK-UP DELAY (FLIP-FLOP)
	MAKE AND BREAK CONTACT OF THERMAL RELAY
	MAKE AND BREAK CONTACT OF LIMIT SWITCH
n On-1	MULTI POSITION SWITCH (GENERAL SYMBOL)
F	MAKE AND BREAK CONTACT OPERATED BY TURNING WITH SPRING RETURN
E	MAKE AND BREAK CONTACT OPERATED BY PUSHING WITH SPRING RETURN
J	MAKE AND BREAK CONTACT OPERATED BY PULLING WITH OR WITHUOT SPRING RETURN

	MUSHROOM PUSH-BUTTON PULSE CONTACT WITH STAY-PUT MAKE CONTACT OPERATED BY STRIKING
	KEY OPERATING DEVICE FOR SELECTORS, PUSH-BUTTON, ETC.
P	PRESSURE SWITCH
ϑ	TEMPERATURE SWITCH
	FLOW SWITCH
	FLOAT SWITCH
	BUCHHOLZ RELAY
	CAM OPERATED DEVICE
	PROXIMITY SWITCH
ϑ	TEMPERATURE SENSOR
	PHOTOELECTRIC SENSOR
	SOLENOID VALVE
	PHOTOCELL

	STEDY OR FLASHING LIGHT
	PLC UNIT
	CONTROL INSTRUMENT
*	ENERGY METER
*	MEASURING INSTRUMENT
	VOLTAGE TRANSFORMER
	CURRENT TRANSFORMER
	ACOUSTIC SIGNALLING DEVICE
	RESISTANCE
	CAPACITOR
M 1~	SINGLE-PHASE ELECTRIC MOTOR
M 3~	THREE-PHASE ELECTRIC MOTOR

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			Contract : IE25072	Rev. 3 Data : 07/07/2025	Date : 17/02/2025
			S/N : IE25072_SE04	Rev. 3 Data : 17/04/2025	Sheet : 2 of : 16

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20

CONDUCTOR COLOR CODING INSIDE CONTROL PANEL

AC AND DC POWER CIRCUIT	BLACK
NEUTRAL CONDUCTOR OF POWER CIRCUIT	LIGHT–BLUE
PROTECTIVE CONDUCTOR (EARTH CONDUCTOR)	YELLOW–GREEN
110/220V AC AUXILIARY CIRCUIT (*)	RED
24V AC AUXILIARY CIRCUIT	WHITE
24V AC AND DC PLC INPUT	GRAY
24V AC PLC OUTPUT	WHITE
24V AC AND DC INTERLOCKING CIRCUIT	ORANGE
24V DC AUXILIARY CIRCUIT AND PLC OUTPUT	DARK–BLUE
INTRINSIC SAFE SIGNAL	BLUE

(*) : RED FOR 110V AND BROWN FOR 220V, IN CASE BOTH VOLTAGE ARE USED

CODE COLOUR

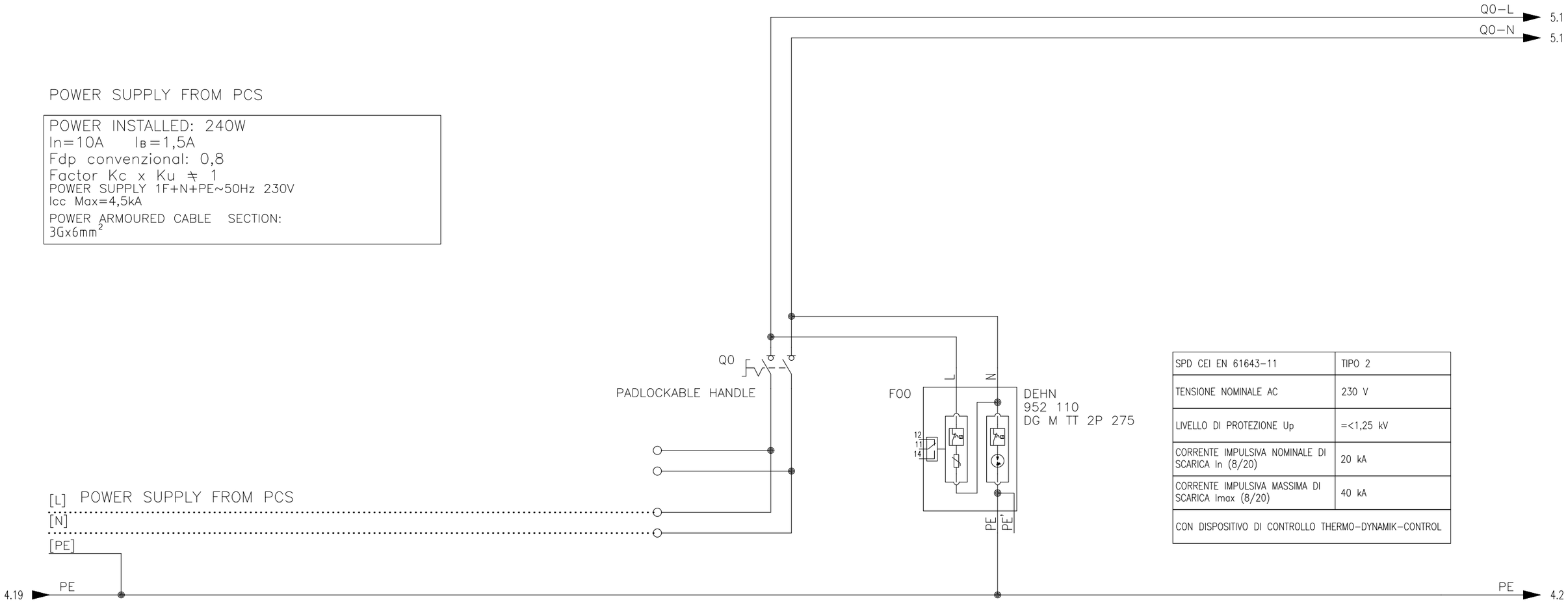
COLOUR	CODE LETTERS
BLACK	BK
BROWN	BN
RED	RD
ORANGE	OG
YELLOW	YE
GREEN	GN
BLUE	BU
GREY	GY
WHITE	WH
SILVER	SR
CLEAR	CL
GREEN AND YELLOW	GNYE

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20

POWER MAIN DISCONNECT SWITCH	FEEDER ID
1F+N+PE~50Hz 230V	RATED POWER Pn–RATED VOLTAGE Vn
	RATED CURRENT

POWER SUPPLY FROM PCS

POWER INSTALLED: 240W
In=10A I_B=1,5A
Fdp convenzional: 0,8
Factor Kc x Ku ≠ 1
POWER SUPPLY 1F+N+PE~50Hz 230V
Icc Max=4,5kA
POWER ARMOURED CABLE SECTION:
3Gx6mm²



SPD CEI EN 61643-11	TIPO 2
TENSIONE NOMINALE AC	230 V
LIVELLO DI PROTEZIONE Up	=<1,25 kV
CORRENTE IMPULSIVA NOMINALE DI SCARICA In (8/20)	20 kA
CORRENTE IMPULSIVA MASSIMA DI SCARICA Imax (8/20)	40 kA
CON DISPOSITIVO DI CONTROLLO THERMO–DYNAMIK–CONTROL	

PROTECTION	TECHNOR PSRC–220	TYPE
		RATED BREAKING CAPACITY Icu
	20A	RATED or THERMAL CURRENT In or Ith
		MAGNETIC RELEASE CARACTERISTICS Im
		RATED R.C. SENSITIVITY Idn
CONDUCTOR		WIRE SECTION
		CABLE TYPE
		CABLE CARRYING LOAD Iz
		CABLE LENGHT
		LINE ID
		WIRE ID ON TERMINAL

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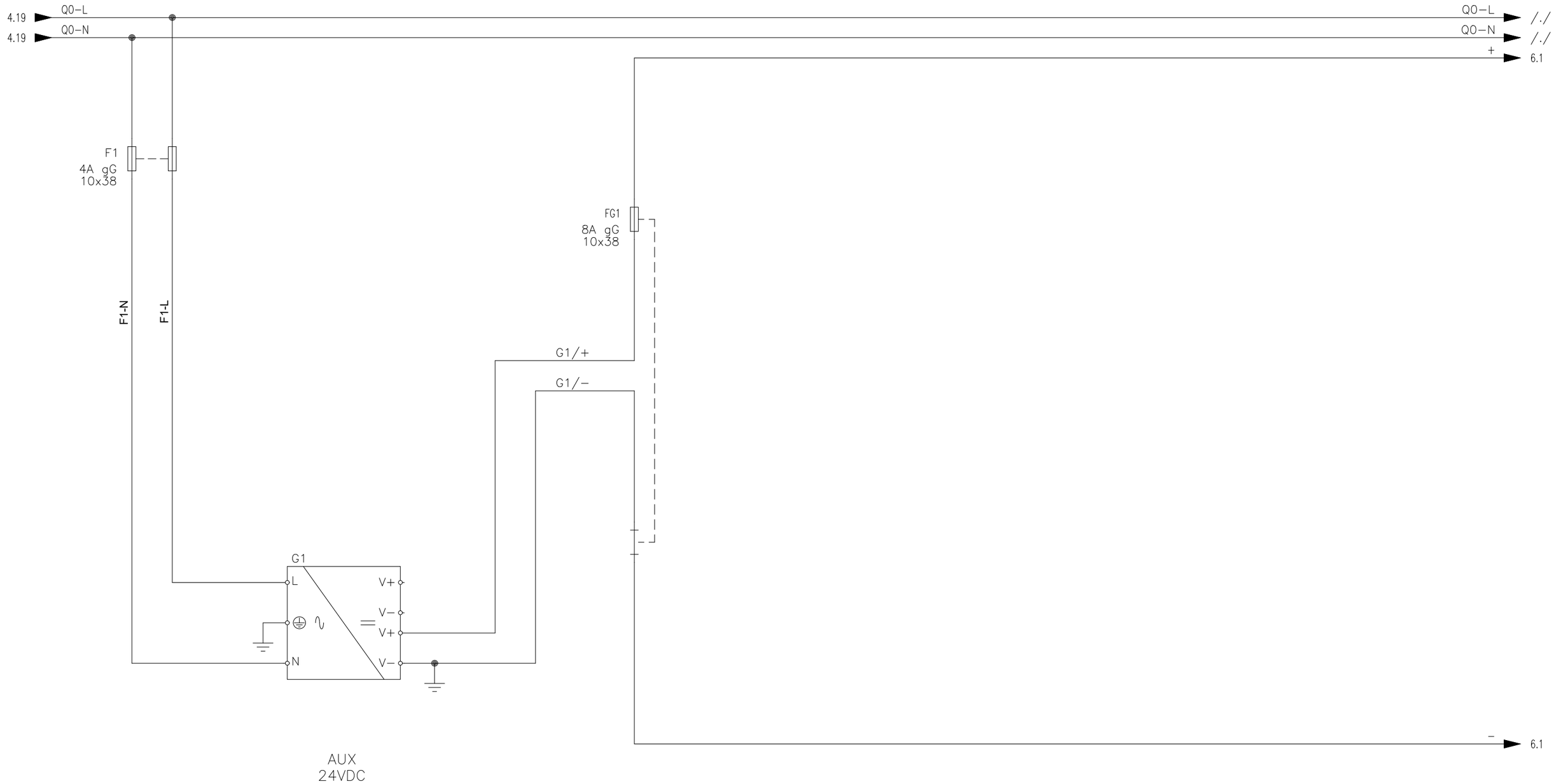
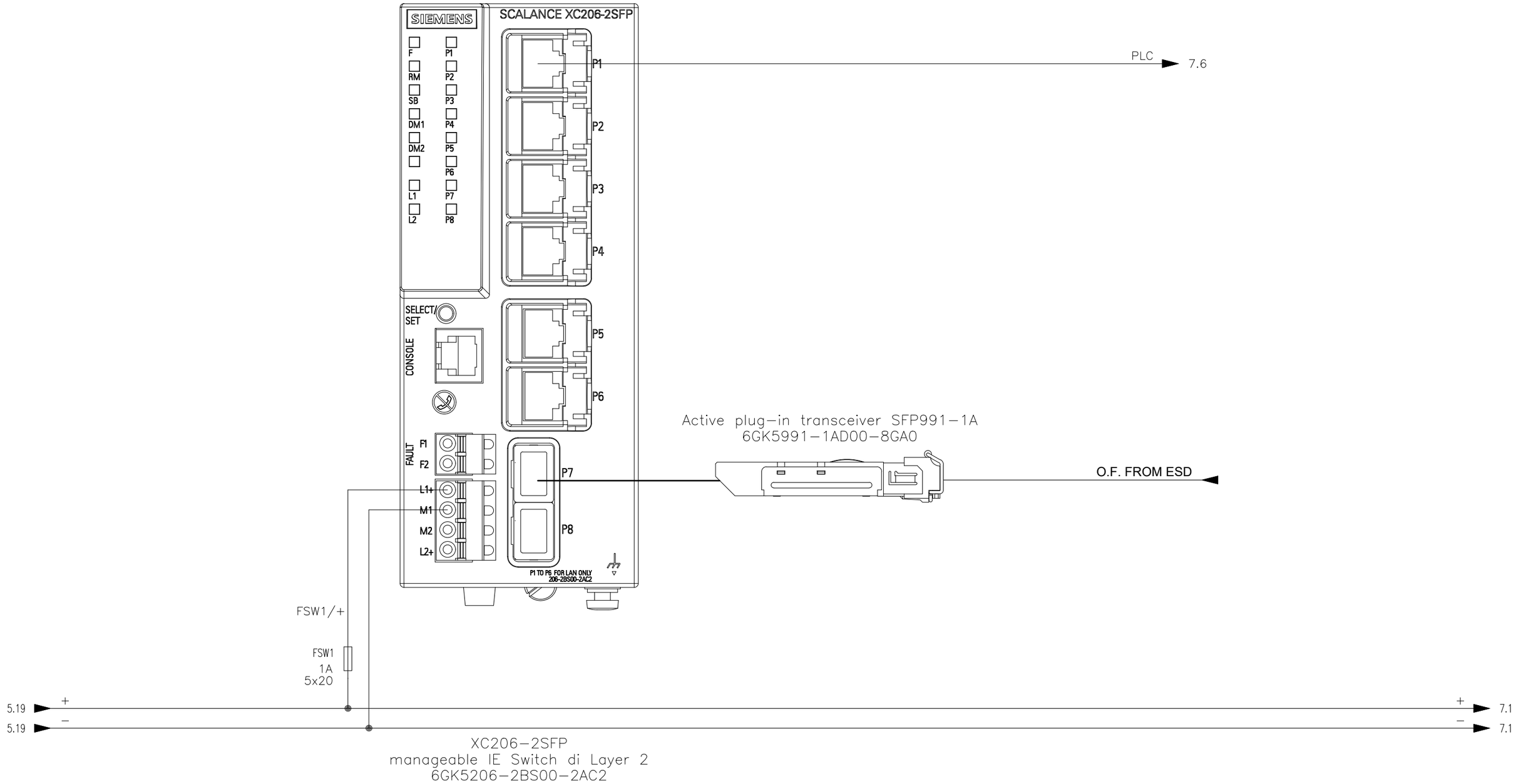
			S/N : IE25072_SE04	Rev. 3 Data : 17/04/2025	Sheet : 4 of : 16

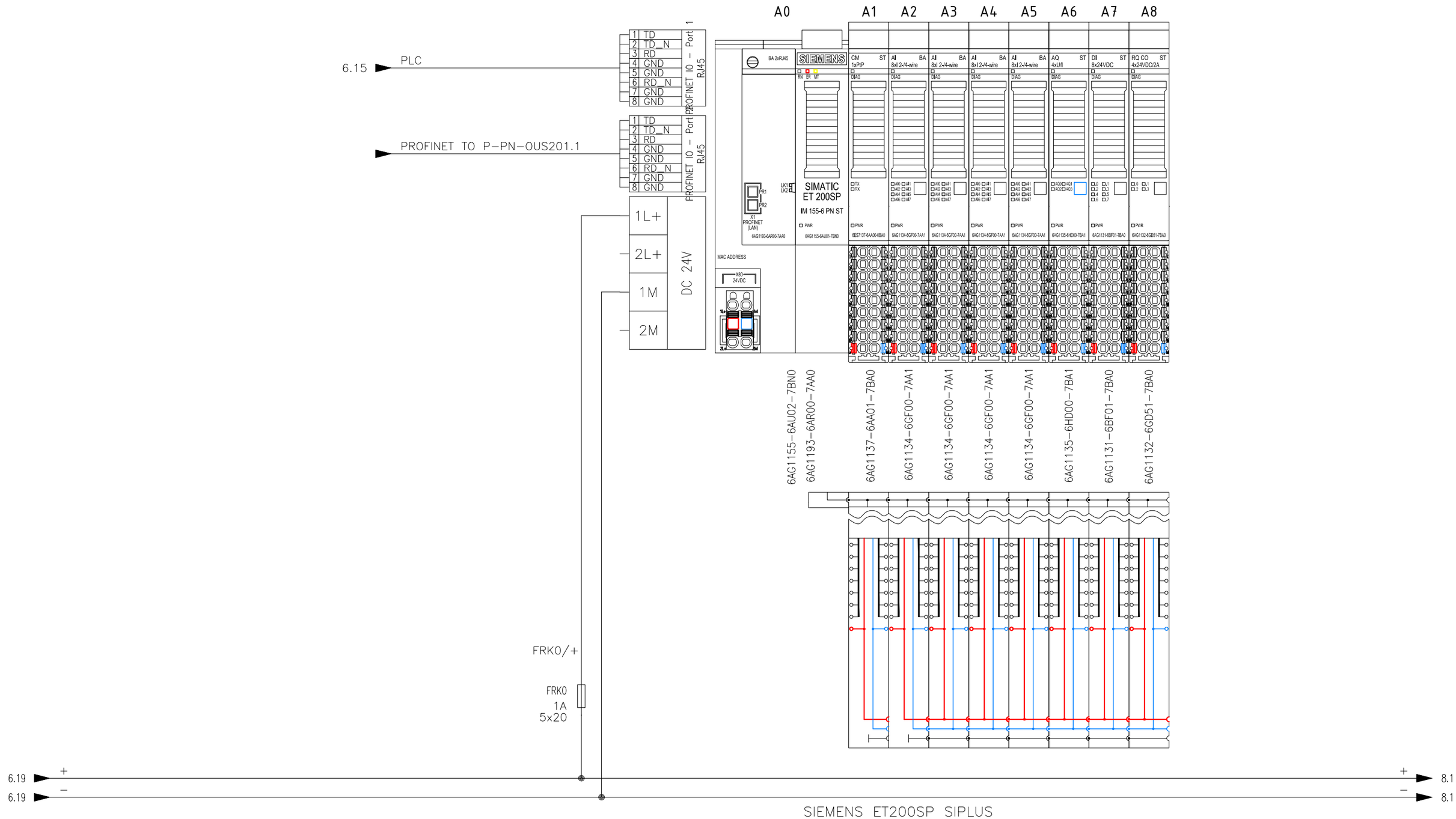


G1: TRIO-PS-2G/1AC/24DC/10 - Power supply unit PHORNIX CONTACT

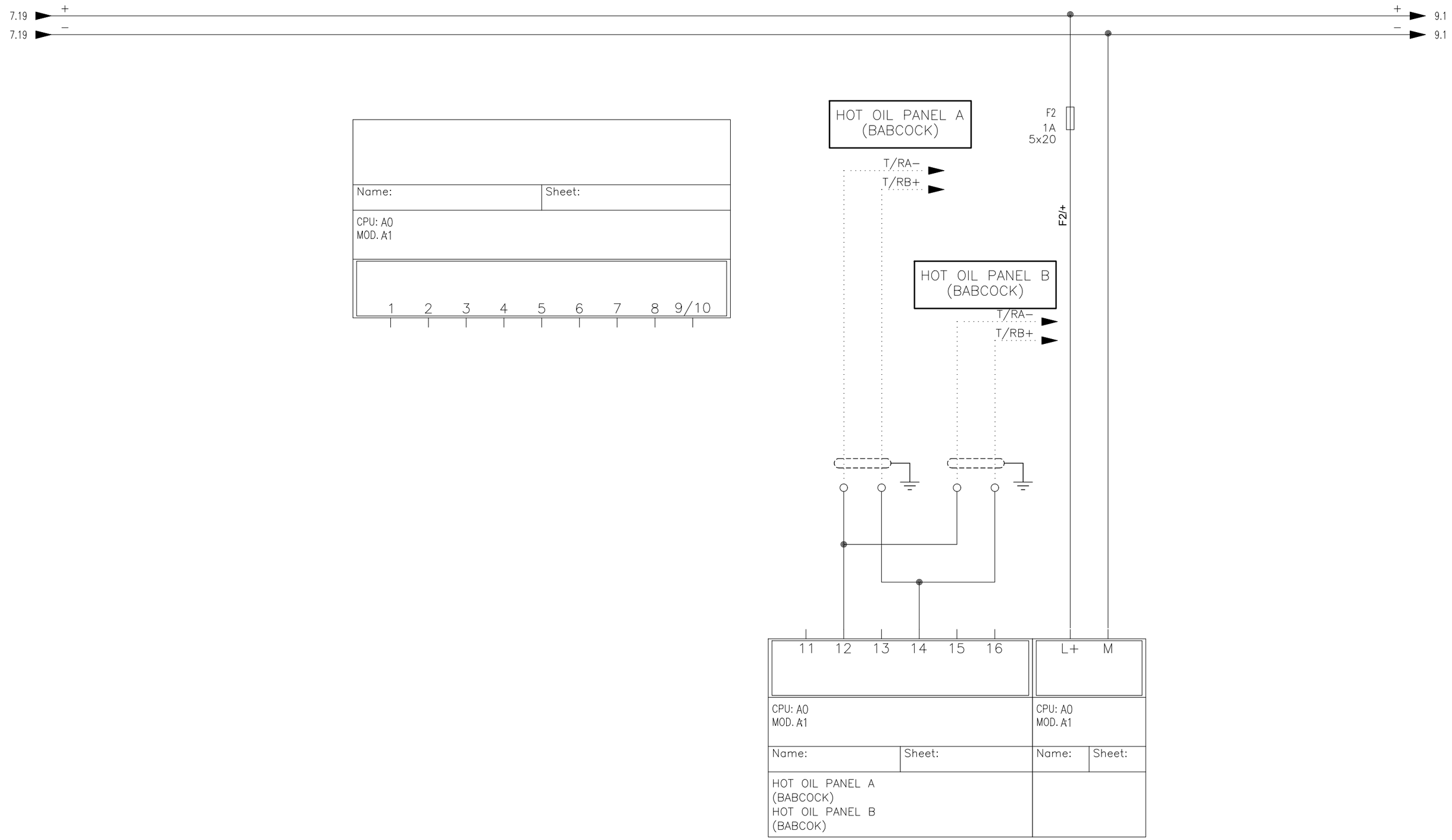
General Contractor :	Manufacture : esa	Project : P-PN-OS412.1 HOT OIL CIRCULATION PUMP SKID	Customer : SOUND ENERGY PLC	Object :		By :	
			Contract : IE25072	Rev. 3	Data : 07/07/2025	Date : 17/02/2025	
			S/N : IE25072_SE04	Rev. 3	Data : 17/04/2025	Sheet : 5	of : 16



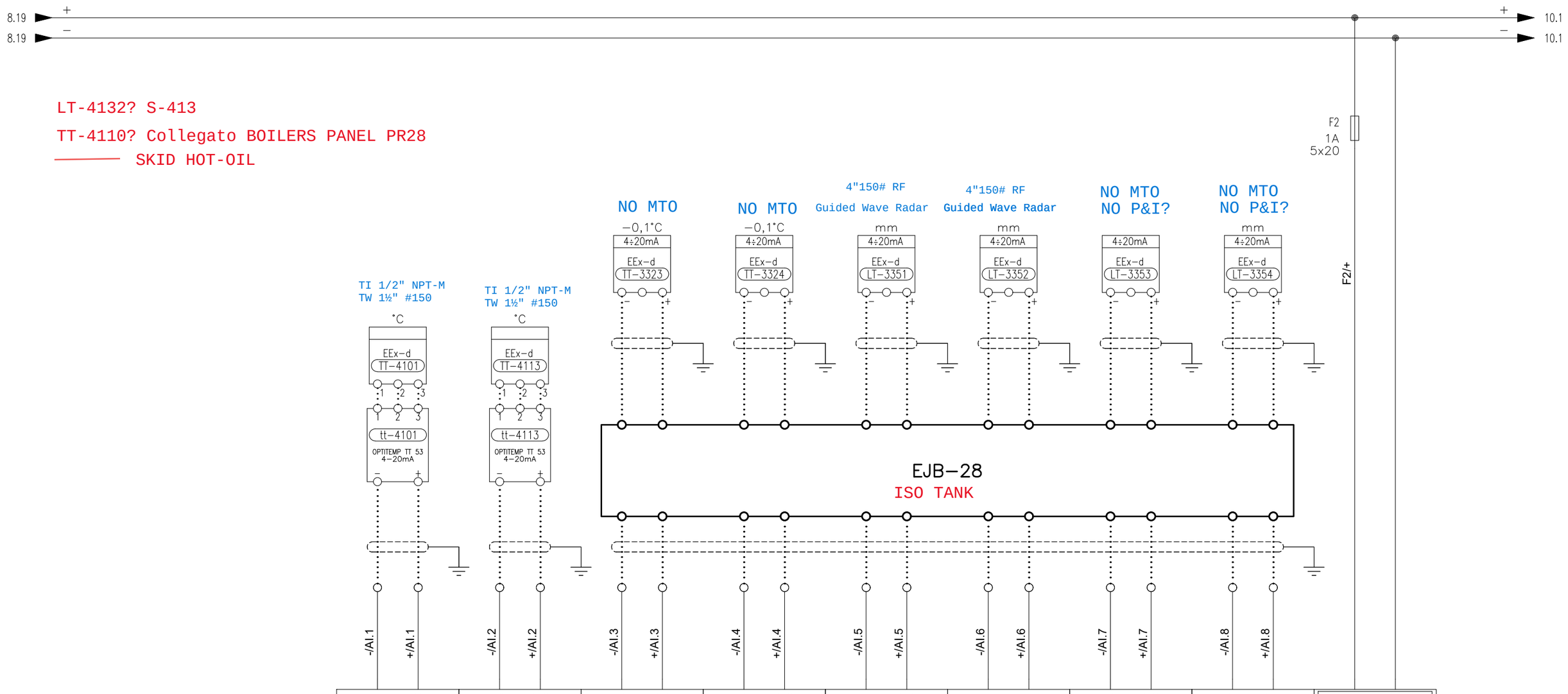
General Contractor :	Manufacture : esa	Project : P-PN-OS412.1 HOT OIL CIRCULATION PUMP SKID	Customer : SOUND ENERGY PLC	Object :		By :	
			Contract : IE25072	Rev. 3	Data : 07/07/2025	Date : 17/02/2025	
			S/N : IE25072_SE04	Rev. 3	Data : 17/04/2025	Sheet : 6	of : 16



General Contractor :	Manufacture : esa	Project : P-PN-OS412.1 HOT OIL CIRCULATION PUMP SKID	Customer : SOUND ENERGY PLC Contract : IE25072 S/N : IE25072_SE04	Object : Rev. 3 Data : 07/07/2025 Rev. 3 Data : 17/04/2025	By : Date : 17/02/2025 Sheet : 7 of : 16

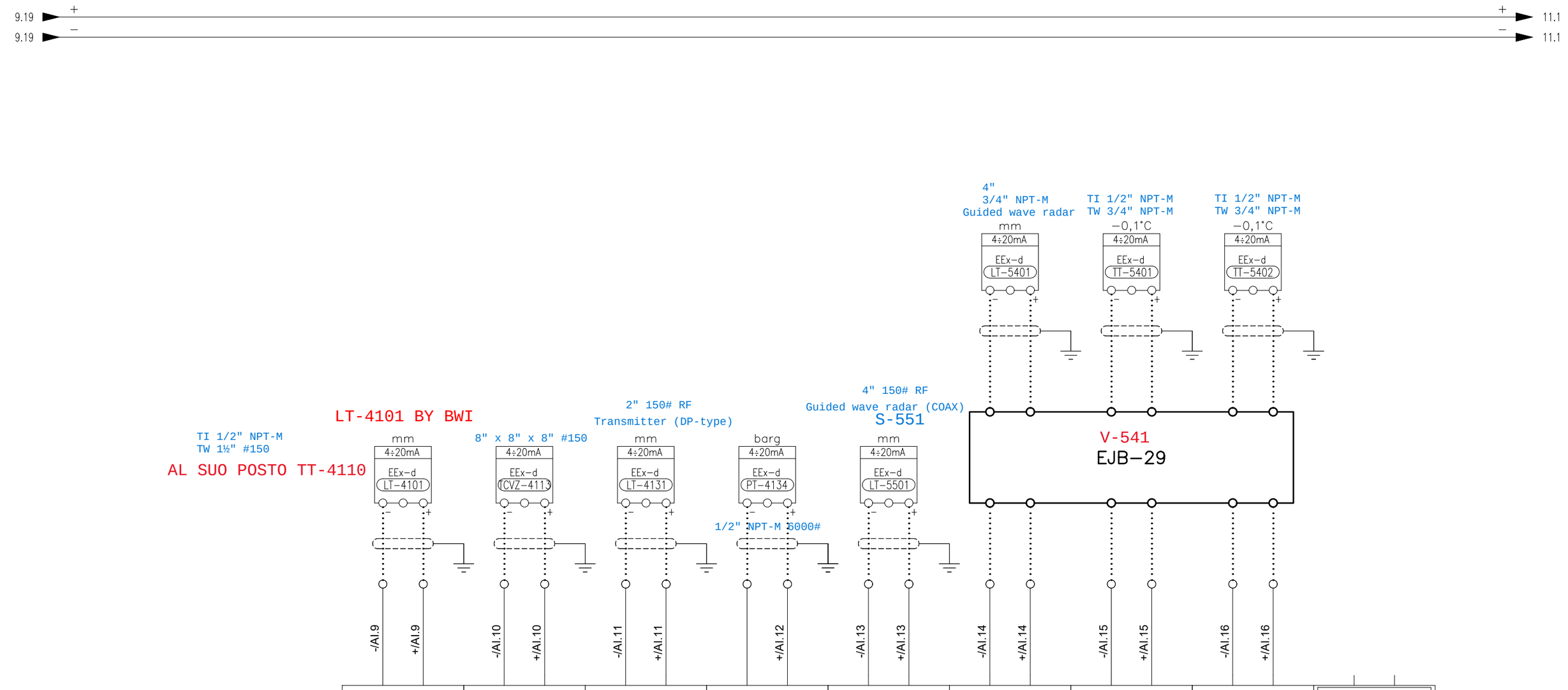


General Contractor :	Manufacture : esa	Project : P−PN−OS412.1 HOT OIL CIRCULATION PUMP SKID	Customer : SOUND ENERGY PLC	Object :	By :
			Contract : IE25072	Rev. 3 Data : 07/07/2025	Date : 17/02/2025
			S/N : IE25072_SE04	Rev. 3 Data : 17/04/2025	Sheet : 8 of : 16



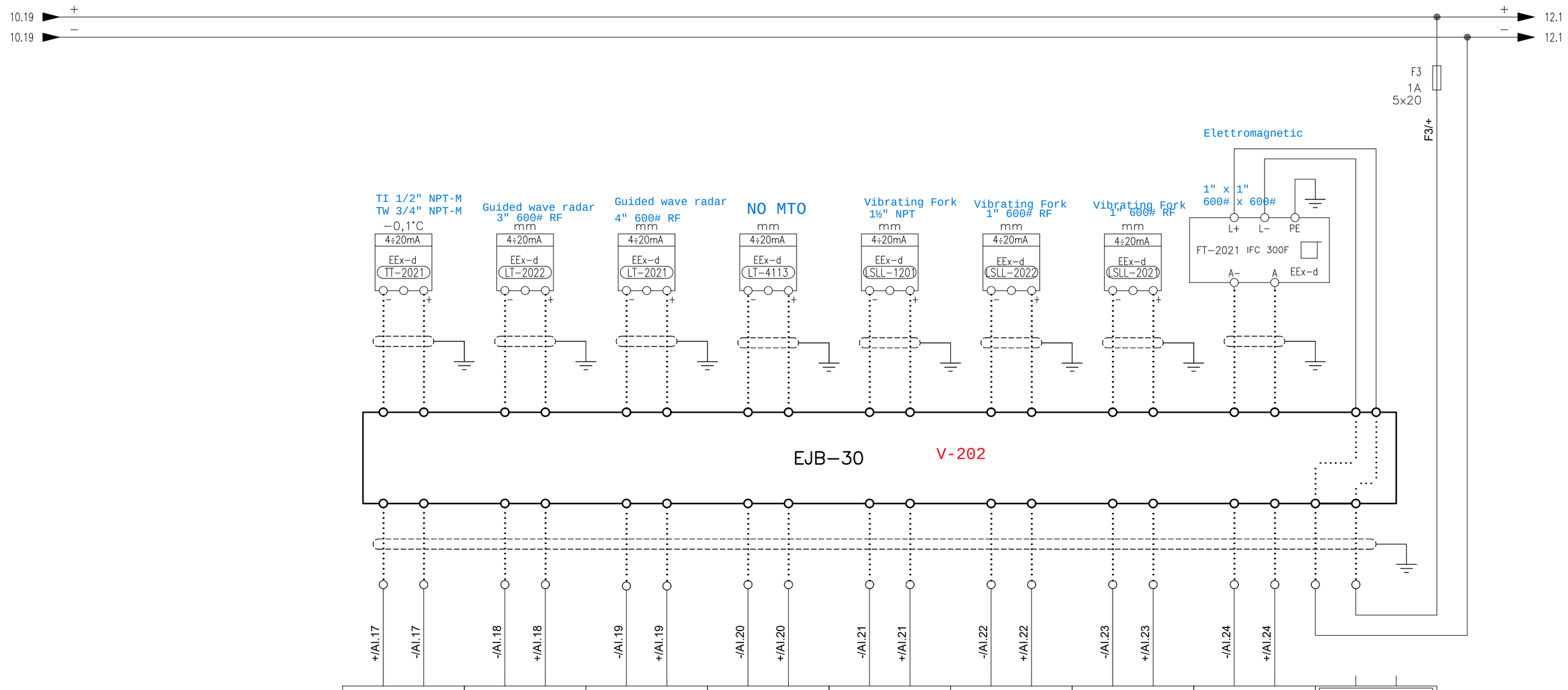
1 9	2 10	3 11	4 12	5 13	6 14	7 15	8 16	L+ M
AI.1	AI.2	AI.3	AI.4	AI.5	AI.6	AI.7	AI.8	
CPU: A0 MOD. A2	CPU: A0 MOD. A2	CPU: A0 MOD. A2	CPU: A0 MOD. A2	CPU: A0 MOD. A2	CPU: A0 MOD. A2	CPU: A0 MOD. A2	CPU: A0 MOD. A2	CPU: A0 MOD. A2
Name: Sheet:	Name: Sheet:	Name: Sheet:	Name: Sheet:	Name: Sheet:	Name: Sheet:	Name: Sheet:	Name: Sheet:	Name: Sheet:
HOT OIL TO B−411A/B TEMPERATURE	HOT OIL TO E−201/E−371/ E−372 TEMPERATURE	V−335 TEMPERATURE	V−336 TEMPERATURE	V−335 LIQUID LEVEL	V−336 FRESH AMINE TANK	V−338−1 EXHAUST AMINE STORAGE TANK	V−338−2 EXHAUST AMINE STORAGE TANK	

General Contractor :	Manufacture : esa	Project : P−PN−OS412.1 HOT OIL CIRCULATION PUMP SKID	Customer : SOUND ENERGY PLC Contract : IE25072 S/N : IE25072_SE04	Object : Rev. 3 Data : 07/07/2025 Rev. 3 Data : 17/04/2025	By : Date : 17/02/2025 Sheet : 9 of : 16



1 9	2 10	3 11	4 12	5 13	6 14	7 15	8 16	L+ M
AI.9	AI.10	AI.11	AI.12	AI.13	AI.14	AI.15	AI.16	
CPU: A0 MOD. A3	CPU: A0 MOD. A3	CPU: A0 MOD. A3	CPU: A0 MOD. A3	CPU: A0 MOD. A3	CPU: A0 MOD. A3	CPU: A0 MOD. A3	CPU: A0 MOD. A3	CPU: A0 MOD. A3
Name: Sheet:	Name: Sheet:	Name: Sheet:	Name: Sheet:	Name: Sheet:	Name: Sheet:	Name: Sheet:	Name: Sheet:	Name: Sheet:
V−412 LIQUID LEVEL	HOT OIL TO E−201 E−371 / E−372 TEMPERATURE CONTROL	S−413 LIQUID LEVEL	S−551 LIQUID LEVEL	S−551 LIQUID LEVEL	V−541 RAW WATER STRORAGE TANK TRASMITTER	V−541 RAW WATER STRORAGE TANK TEMPERATURE	TEMPERATURE TRASMITTER	

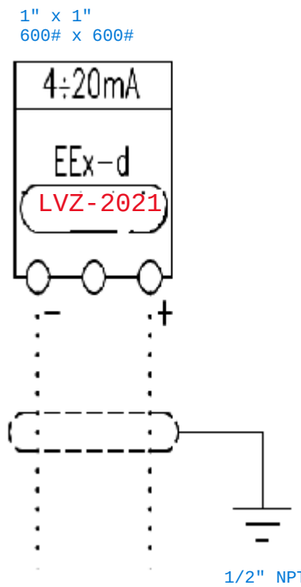
General Contractor :	Manufacture : esa elettromeccanica.com	Project : P−PN−OS412.1 HOT OIL CIRCULATION PUMP SKID	Customer : SOUND ENERGY PLC Contract : IE25072 S/N : IE25072_SE04	Object : Rev. 3 Data : 07/07/2025 Rev. 3 Data : 17/04/2025	By : Date : 17/02/2025 Sheet : 10 of : 16



1 9	2 10	3 11	4 12	5 13	6 14	7 15	8 16	L+ M
AI.17	AI.18	AI.19	AI.20	AI.21	AI.22	AI.23	AI.24	
CPU: A0 MOD. A4	CPU: A0 MOD. A4	CPU: A0 MOD. A4	CPU: A0 MOD. A4	CPU: A0 MOD. A4	CPU: A0 MOD. A4	CPU: A0 MOD. A4	CPU: A0 MOD. A4	CPU: A0 MOD. A4
Name: Sheet:	Name: Sheet:	Name: Sheet:	Name: Sheet:	Name: Sheet:	Name: Sheet:	Name: Sheet:	Name: Sheet:	Name: Sheet:
V−202 GAS OUTLET TEMPERATURE	V−202 HEAVY LEVEL TRASMITTER	V−202 LIGHT LEVEL TRASMITTER	TK−411 HOT OIL DIESEL TANK	120−V001 LIQUID LOW LEVEL SWITCH	V−202 LOW LEVEL SWITCH	V−202 LIGHT HIGH LEVEL SWITCH	V−202 OUTLET WATER GAS FLOW	

General Contractor :	Manufacture : esa elettromeccanica.com	Project : P−PN−OS412.1 HOT OIL CIRCULATION PUMP SKID	Customer : SOUND ENERGY PLC Contract : IE25072 S/N : IE25072_SE04	Object : Rev. 3 Data : 07/07/2025 Rev. 3 Data : 17/04/2025
By :	Date : 17/02/2025	Sheet : 11 of : 16		

11.19 + → 13.1 +
11.19 − → 13.1 −



-/AI.25	+/AI.25	-/AI.26	+/AI.26	-/AI.27	+/AI.27	-/AI.28	+/AI.28	-/AI.29	+/AI.29	-/AI.30	+/AI.30	-/AI.31	+/AI.31	-/AI.32	+/AI.32	L+	M
1	9	2	10	3	11	4	12	5	13	6	14	7	15	8	16		
AI.25		AI.26		AI.27		AI.28		AI.29		AI.30		AI.31		AI.32			
CPU: AO MOD. A5		CPU: AO MOD. A5		CPU: AO MOD. A5		CPU: AO MOD. A5		CPU: AO MOD. A5		CPU: AO MOD. A5		CPU: AO MOD. A5		CPU: AO MOD. A5		CPU: AO MOD. A5	
Name:	Sheet:	Name:	Sheet:	Name:	Sheet:	Name:	Sheet:	Name:	Sheet:	Name:	Sheet:	Name:	Sheet:	Name:	Sheet:	Name:	Sheet:
SPARE LVZ-2021 V-202 LIGHT LIQUID LEVEL CONTROL VALVE		SPARE PT-6001		SPARE		SPARE		SPARE		SPARE		SPARE		SPARE			
EJB-31		EJB-31															

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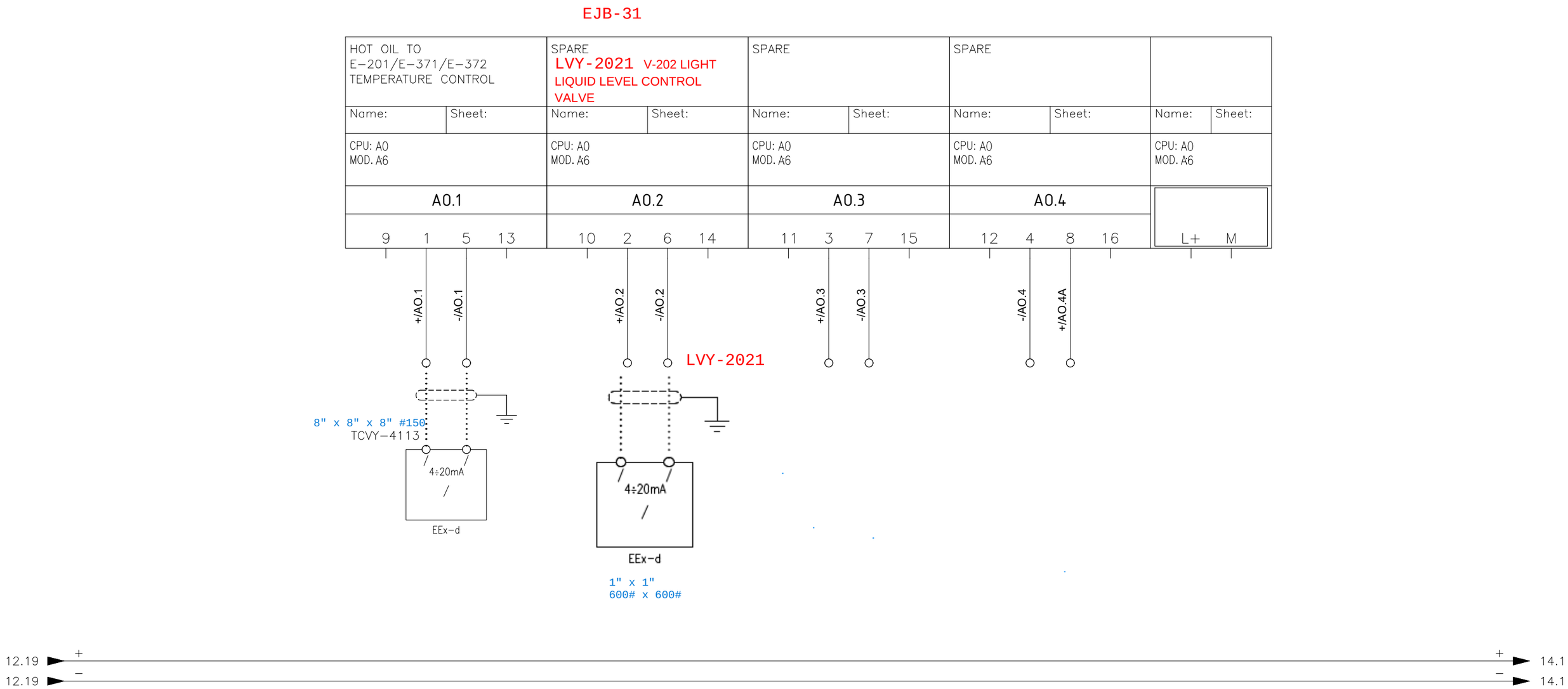

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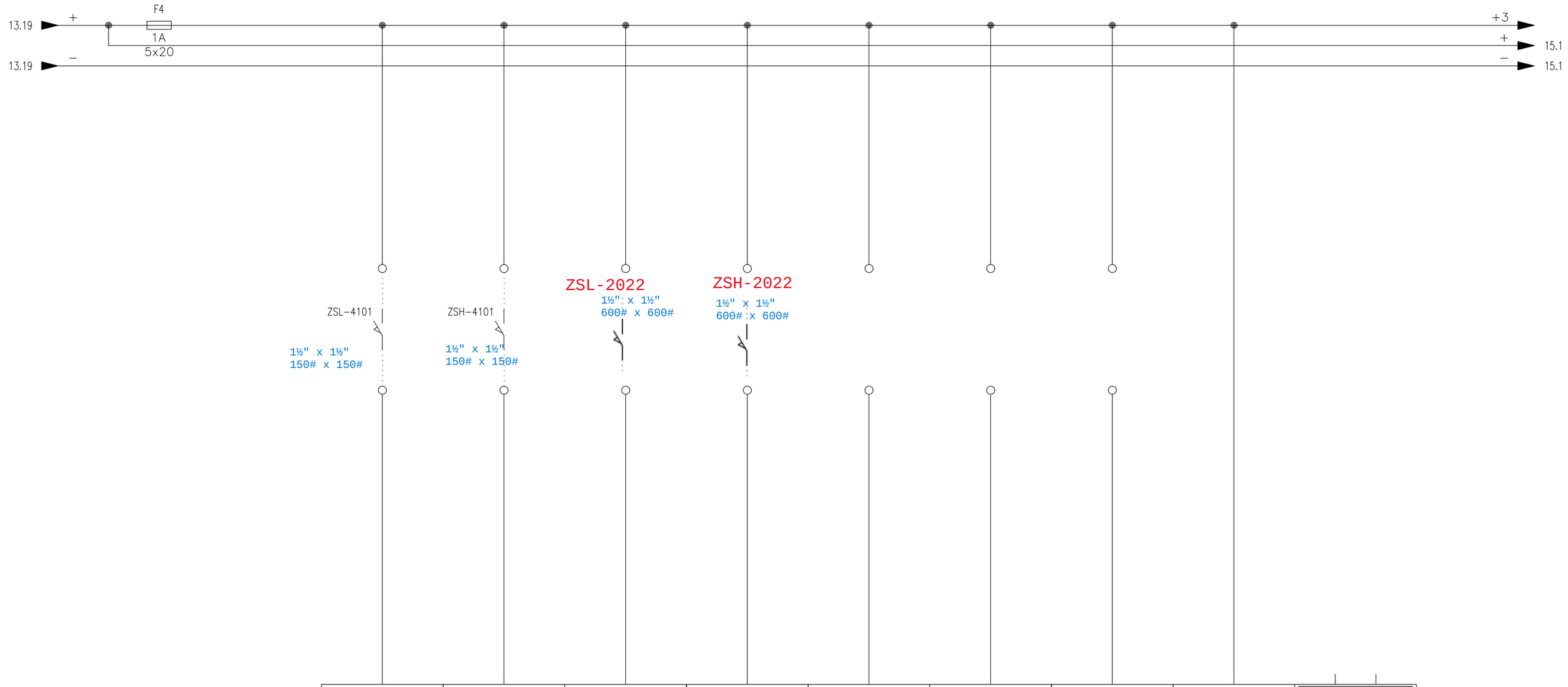

1	2	3	4	5	6	7	8	L+ M
DI.1	DI.2	DI.3	DI.4	DI.5	DI.6	DI.7	DI.8	
CPU: A0 MOD. A7	CPU: A0 MOD. A7	CPU: A0 MOD. A7	CPU: A0 MOD. A7	CPU: A0 MOD. A7	CPU: A0 MOD. A7	CPU: A0 MOD. A7	CPU: A0 MOD. A7	CPU: A0 MOD. A7
Name: Sheet:	Name: Sheet:	Name: Sheet:	Name: Sheet:	Name: Sheet:	Name: Sheet:	Name: Sheet:	Name: Sheet:	Name: Sheet:
XV–4101 CLOSED POSITION INDICATION	XV–4101 OPEN POSITION INDICATION	SPARE LV-2022 CLOSED POSITION INDICATION EJB-31	SPARE LV-2022 OPEN POSITION INDICATION EJB-31	SPARE	SPARE	SPARE	CONTROL FUSE	

General Contractor :	Manufacture : esa	Project : P–PN–OS412.1 HOT OIL CIRCULATION PUMP SKID	Customer : SOUND ENERGY PLC Exploration & Production	Object :	By :
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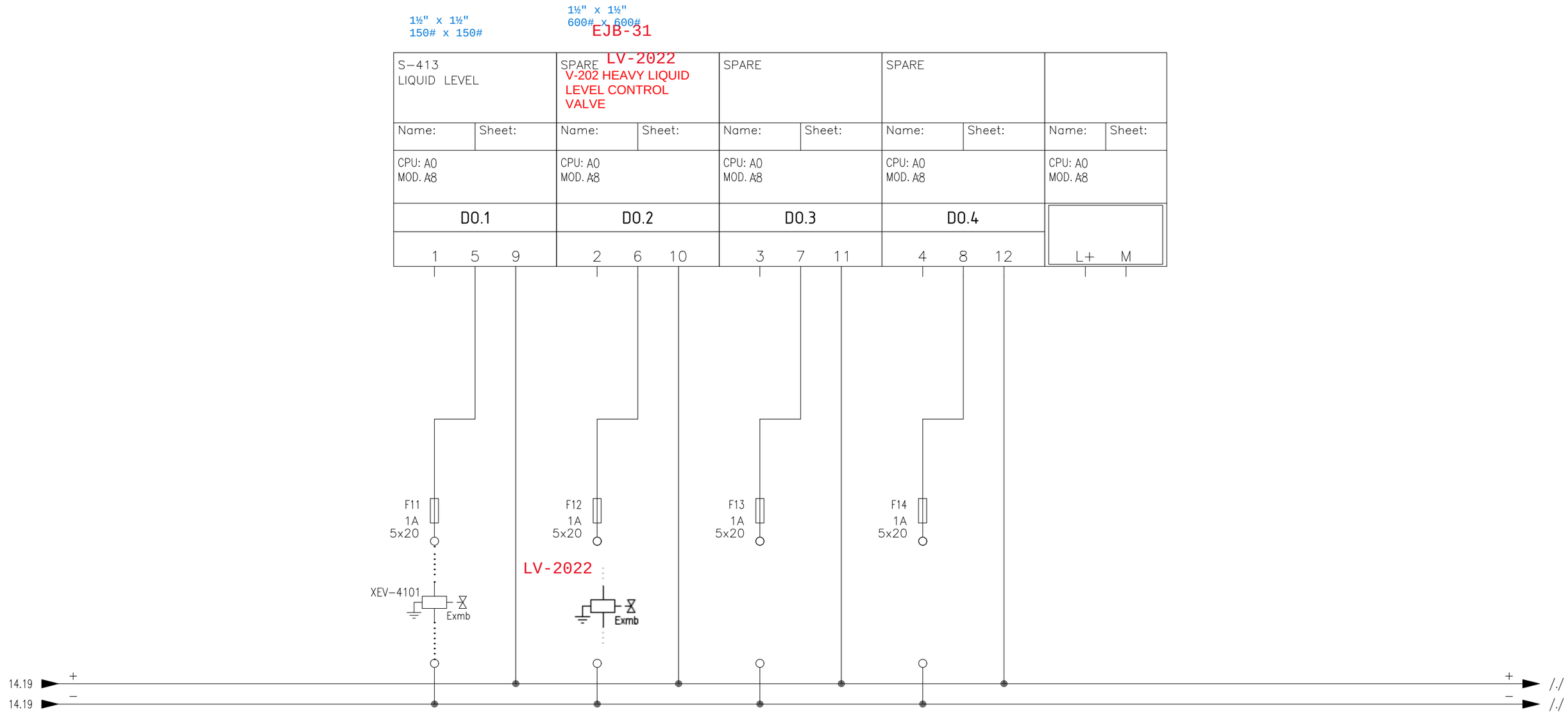

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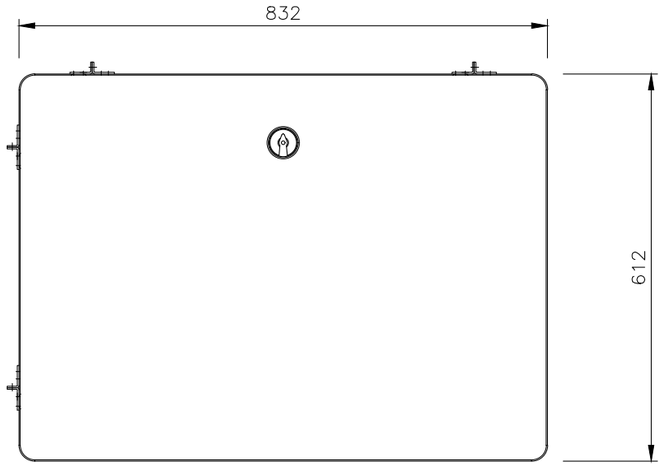
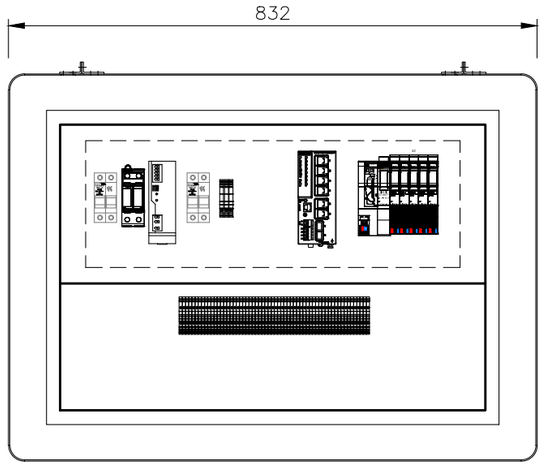
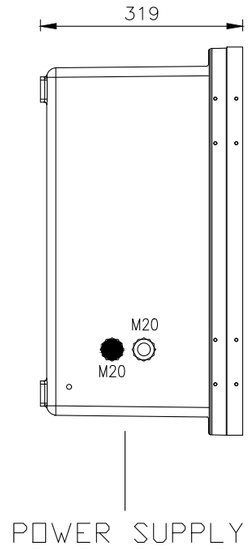
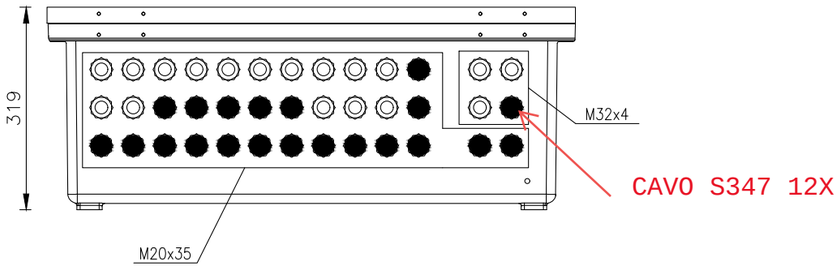
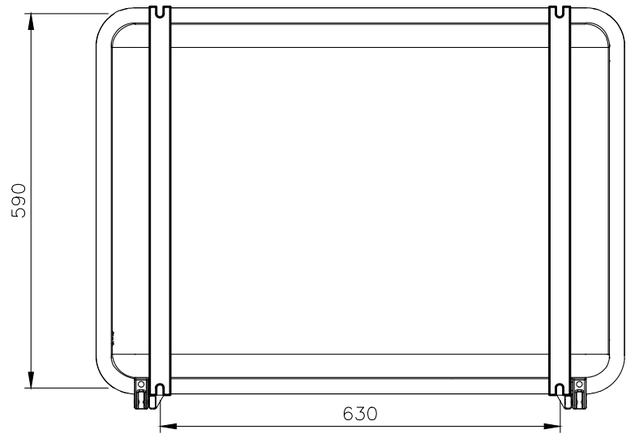
NOTE
- Type : EJB-G
- Ex-d [ia] IIB T6 IP65 ATEX -20/+50°C
- Installation area is zone 2 IIA T1
- Power dissipation 50W











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			S/N : IE25072_SE04	Rev. 3	Data : 17/04/2025	Sheet : 16	of : 16